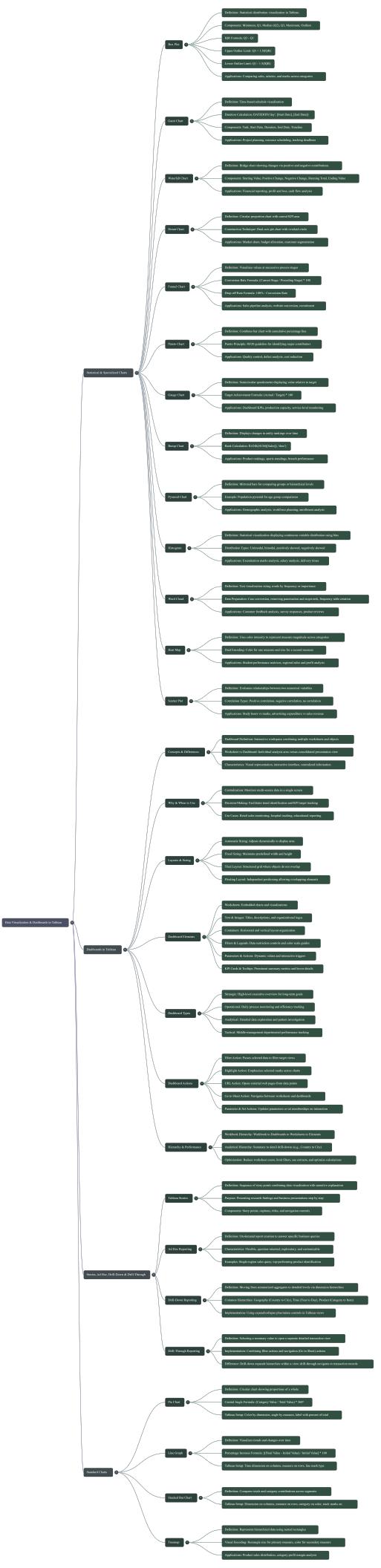

Data Visualization & Dashboards in Tableau
Statistical & Specialized Charts
Box Plot
Definition: Statistical distribution visualization in Tableau
Components: Minimum, Q1, Median (Q2), Q3, Maximum, Outliers
IQR Formula: Q3 - Q1
Upper Outlier Limit: Q3 + 1.5(IQR)
Lower Outlier Limit: Q1 - 1.5(IQR)
Applications: Comparing sales, salaries, and marks across categories
Gantt Chart
Definition: Time-based schedule visualization
Duration Calculation: DATEDIFF('day', [Start Date], [End Date])
Components: Task, Start Date, Duration, End Date, Timeline
Applications: Project planning, resource scheduling, tracking deadlines
Waterfall Chart
Definition: Bridge chart showing changes via positive and negative contributions
Components: Starting Value, Positive Changes, Negative Changes, Running Total, Ending Value
Applications: Financial reporting, profit and loss, cash flow analysis
Donut Chart
Definition: Circular proportion chart with central KPI area
Construction Technique: Dual-axis pie chart with overlaid circle
Applications: Market share, budget allocation, customer segmentation
Funnel Chart
Definition: Visualizes values at successive process stages
Conversion Rate Formula: (Current Stage / Preceding Stage) * 100
Drop-off Rate Formula: 100% - Conversion Rate
Applications: Sales pipeline analysis, website conversion, recruitment
Pareto Chart
Definition: Combines bar chart with cumulative percentage line
Pareto Principle: 80/20 guideline for identifying major contributors
Applications: Quality control, defect analysis, cost reduction
Gauge Chart
Definition: Semicircular speedometer displaying value relative to target
Target Achievement Formula: (Actual / Target) * 100
Applications: Dashboard KPIs, production capacity, service-level monitoring
Bump Chart
Definition: Displays changes in entity rankings over time
Rank Calculation: RANK(SUM([Sales]), 'desc')
Applications: Product rankings, sports standings, branch performance
Pyramid Chart
Definition: Mirrored bars for comparing groups at hierarchical levels
Example: Population pyramid for age group comparisons
Applications: Demographic analysis, workforce planning, enrollment analysis
Histogram
Definition: Statistical visualization displaying continuous variable distribution using bins
Distribution Types: Unimodal, bimodal, positively skewed, negatively skewed
Applications: Examination marks analysis, salary analysis, delivery times
Word Cloud
Definition: Text visualization sizing words by frequency or importance
Data Preparation: Case conversion, removing punctuation and stopwords, frequency table creation
Applications: Customer feedback analysis, survey responses, product reviews
Heat Map
Definition: Uses color intensity to represent measure magnitude across categories
Dual Encoding: Color for one measure and size for a second measure
Applications: Student performance matrices, regional sales and profit analysis
Scatter Plot
Definition: Evaluates relationships between two numerical variables
Correlation Types: Positive correlation, negative correlation, no correlation
Applications: Study hours vs marks, advertising expenditure vs sales revenue
Dashboards in Tableau
Concepts & Differences
Dashboard Definition: Interactive workspace combining multiple worksheets and objects
Worksheet vs Dashboard: Individual analysis area versus consolidated presentation view
Characteristics: Visual representation, interactive interface, centralized information
Why & When to Use
Centralization: Monitors multi-source data in a single screen
Decision-Making: Facilitates trend identification and KPI target tracking
Use Cases: Retail sales monitoring, hospital tracking, educational reporting
Layouts & Sizing
Automatic Sizing: Adjusts dynamically to display area
Fixed Sizing: Maintains predefined width and height
Tiled Layout: Structured grid where objects do not overlap
Floating Layout: Independent positioning allowing overlapping elements
Dashboard Elements
Worksheets: Embedded charts and visualizations
Text & Images: Titles, descriptions, and organizational logos
Containers: Horizontal and vertical layout organization
Filters & Legends: Data restriction controls and color scale guides
Parameters & Actions: Dynamic values and interactive triggers
KPI Cards & Tooltips: Prominent summary statistics and hover details
Dashboard Types
Strategic: High-level executive overviews for long-term goals
Operational: Daily process monitoring and efficiency tracking
Analytical: Detailed data exploration and pattern investigation
Tactical: Middle-management departmental performance tracking
Dashboard Actions
Filter Action: Passes selected data to filter target views
Highlight Action: Emphasizes selected marks across charts
URL Action: Opens external web pages from data points
Go to Sheet Action: Navigates between worksheets and dashboards
Parameter & Set Actions: Updates parameters or set memberships on interaction
Hierarchy & Performance
Worksheet Hierarchy: Worksheets vs Dashboards vs Workbooks vs Extracts
Analytical Hierarchy: Summary to detail drill-downs (e.g., Country to City)
Optimization: Reduce worksheet count, limit filters, use extracts, and optimize calculations
Stories, Ad Hoc, Drill-Down & Drill-Through
Tableau Stories
Definition: Sequence of story points combining data visualizations with narrative explanation
Purpose: Presenting research findings and business presentations step by step
Components: Story points, captions, titles, and navigation controls
Ad Hoc Reporting
Definition: On-demand report creation to answer specific business queries
Characteristics: Flexible, question-oriented, exploratory, and customizable
Examples: Single-region sales query, top-performing product identification
Drill-Down Reporting
Definition: Moving from summarized aggregates to detailed levels via dimension hierarchies
Common Hierarchies: Geography (Country to City), Time (Year to Day), Product (Category to Item)
Implementation: Using expand/collapse plus hierarchy controls in Tableau views
Drill-Through Reporting
Definition: Selecting a summary value to open a separate detailed transaction view
Implementation: Customizing filter actions and navigation (Go to Sheet) actions
Difference: Drill-down expands hierarchies within a view; drill-through navigates to transaction records
Standard Charts
Pie Chart
Definition: Circular chart showing proportions of a whole
Central Angle Formula: (Category Value / Total Values) * 360°
Tableau Setup: Color by dimension, angle by measure, label with percent of total
Line Graph
Definition: Visualizes trends and changes over time
Percentage Increase Formula: [(Final Value - Initial Value) / Initial Value] * 100
Tableau Setup: Time dimension on columns, measure on rows, line mark type
Stacked Bar Chart
Definition: Compares totals and category contributions across segments
Tableau Setup: Dimensions on columns, measure on rows, category on color, mark marks on
Treemap
Definition: Represents hierarchical data using nested rectangles
Visual Encoding: Rectangle size for primary measure, color for secondary measure
Applications: Product sales distribution, category-profit margin analysis